

Recurrent mutation patterns in antimicrobial-peptide evolution experiments: a preregistered, descriptive audit with an unmet discovery gate

mega27-09e (ResistPep), status report of 8 October 2026

Repository: github.com/uditakankananonononono/mega27-09e-resistpep

Abstract

This lane set out to predict resistance emergence for antimicrobial peptides (AMPs) from public serial-passage data and to nominate named, evolutionarily robust candidates. The locked predictive task was not supported by the data: on 9 units the leave-one-study-out Spearman was -0.547 (95% bootstrap CI -0.915 to 0.116 ; permutation $p = 0.924$), and on the enlarged 21-unit base it was 0.112 (CI -0.275 to 0.490 ; $p = 0.281$). No candidate peptide is nominated. The work then moved, under dated preregistration addenda A4 to A15, to a descriptive audit of mutation recurrence in published AMP evolution experiments. One descriptive observation survives the locked *E. coli* analysis: in 20 lines of Spohn et al. 2019 that carry no sbmA deletion, four (AMP, gene) pairs show Tier A hits in at least three same-AMP lines (basS in CAP18, lptC and wzzE in LL-37, basR in HBD3). Zero of 100,000 draws from the specified length-weighted redraw null reached four (a Monte Carlo tail estimate below 10^{-5}), but that null omits mutation hotspots, linkage and dependence among lines, so this is not a calibrated probability under a realistic null. A later margin-conditional permutation (A16) gives $p = 0.0025$ but its linkage-collapsed sensitivity does not stand out ($p = 0.078$). A second-organism analysis in *S. maltophilia* (Blanco et al. 2020) is exploratory because its gene mapping was chosen after seeing the recurring genes, and it is not a replication. Lofton et al. 2013 gives external experimental evidence, partial and not an independent replication (the study had already been mined by this lane), that pmrB, phoP and waaY mutations recur under LL-37 passaging in *Salmonella*. Several earlier readings did not survive: sbmA recurrence does not exceed a mutation-opportunity null ($p = 0.339$), the sbmA block deletions are breakpoint driven, waaY/phoP concordance failed its locked rule, and an interim claim of waaY dependence on sbmA-deletion lines was withdrawn after a denominator error was corrected. The 09d discovery gate is **unmet**: no independent exact-peptide mass-spectrometry or PubMed record satisfies the locked criteria for the surviving genes. Closing it requires a targeted functional test. This document makes no discovery claim. Its real body length is stated in Section 8, and the repository's 50-page requirement is **unmet**.

1 Introduction and scope

The lane (MEGA-PROGRAM-27, item 9, project 5) was preregistered on 26 September 2026 with a de-risked scope: predict measured resistance-emergence outcomes in public serial-passage datasets and build a robustness-proxy ranking, then nominate named candidates that are predicted antimicrobial and evolutionarily robust across independent evidence layers. The house rules require real public data only, locked splits before outcomes, preserved negatives that are never terminal, a dated addendum before every new evaluation, in-document disclosure of deviations, and an independent-evidence gate (09d) before any named discovery.

This report states what the locked analyses did and did not show. Every number is taken from a result file committed in the repository, each analysis having a lock commit that precedes its execution. Where an earlier number was wrong, the original is kept and the correction is stated beside it. Third-party advice (including one ChatGPT judge round on the A5 finding) was treated as untrusted and used only where re-derived.

2 Data and preregistration framework

2.1 Datasets

Pinned files carry SHA-256 hashes in `data/SHA256SUMS.txt`. The evidence base is:

- Lofton et al. 2013 (cited as “Prabhu 2013” in the A1 addendum; same PLoS One study, PMC3720879; *S. typhimurium* LT2; LL-37, CNY100HL, wheat germ histones; 6 lineages per treatment).
- Antunes 2024 (*P. aeruginosa* PA14; 6 single AMPs and 3 mixtures; 6 evolved and 6 control lines each; Zenodo 11209304).
- Maron 2025 (*S. aureus*; temporin, melittin, pexiganan).
- Spohn et al. 2019 Supplementary Data 5: 38 whole-genome-sequenced AMP-adapted *E. coli* K-12 BW25113 lines. Of these, 28 lines carry the six adopted AMPs in the ledger (BAC5 4, CAP18 5, HBD3 4, LL-37 5, PEX 5, PR39 5); PROA and PXB lines (5 each) are not adopted.
- Blanco et al. 2020 (*S. maltophilia* D457; PR-39 and LL-37; 8 evolved lines each) and Bac7 2023 (*E. coli* MDR 1057; 3 strains).

A source-quality audit found 45 source cells (19 exact, 12 ranges, 8 right-censored, 6 missing). These are nested lineages within treatments and studies, not 45 independent peptide labels. Nine distinct plain single sequences have resistance-emergence outcomes (LL-37, CNY100HL, melittin, pexiganan, cecropin P1, PA-13, SLM1, SLM3, temporin), with melittin and pexiganan appearing in two organisms.

2.2 Mutation ledger and evidence tiers

The A5 ledger holds 405 records (383 Spohn, 16 Blanco evolved, 3 Blanco control, 3 Bac7). A6 defined Tier A events (point mutation, insertion, short or local deletion: direct disruption) and Tier B (membership in a large deletion only). Coordinates from Spohn are BW25113 (CP009273.1) and differ from MG1655 by about 6.7 kb near *sbmA*, so MG1655 is used only by gene name.

2.3 Preregistration and deviations

Each analysis below was locked in a dated addendum committed and pushed before execution. Deviations are disclosed in the result documents and summarised in Table 3. Prior exposure of the analyst to the relevant numbers is stated in each lock.

3 Results

3.1 The locked predictive task is not supported (A1, A3)

The locked primary assumed the public sources support a sequence-level predictive task. A1 disclosed that they do not: labels are lineage-level MIC fold-changes nested within treatment and study, and AUROC on nine sequences is not meaningful. On the nine-unit base (7 folds) the full-descriptor ridge model had leave-one-study-out Spearman -0.547 (with a lipid flag -0.369); the two-feature baseline was -0.343 , the k-mer baseline -0.623 and the global median -0.898 . The bootstrap 95% CI was $[-0.915, 0.116]$ and the permutation null (10,000 draws, seed 1001) gave $p = 0.924$. The BEAT rule (exceed every baseline and a CI excluding 0) failed on both counts. After literature mining under A2, A3 enlarged the base to 21 units with Spohn 2019; the model Spearman was 0.112 (CI $[-0.275, 0.490]$, $p = 0.281$). Neither result supports a predictive claim, and the claim boundary stands: no prediction of resistance for unseen peptides. Mixture-versus-single contrast and cross-resistance asymmetry were retained as descriptive secondaries only (random mixtures net 0.5 to 1.5 $\log_2$ versus single sequences 0.5 to 3.5 $\log_2$ in Antunes 2024).

3.2 Blanco 2020 interval description (A4)

Sixteen evolved endpoints and four reported controls were extracted; all 29 table rows matched an independent text extraction. The median net $\log_2$ MIC interval was $(1, \infty)$ for LL-37 (5 of 8 endpoints right-censored) and $(3, \infty)$ for PR-39 (8 of 8). Intervals are finite-sample bounds, not confidence intervals; different species, media and assays preclude a generalization claim. The Spohn A3 point for LL-37 (4.3502) lies inside the Blanco interval and for PR-39 (1.2925) lies below it.

3.3 Cross-study recurrence and event tiers (A5, A6)

Across the three studies, *sbmA* is the only gene hit in at least two: Spohn PR39 (all 5 sequenced PR39 lines, by short deletion in one line and large deletions in four; the A5 text reported “5/10” using a 10-line denominator that Section 3.8 shows to be wrong) and Bac7 2023 (2 of 3 strains). Under the judge-adopted wording this is “parallel recurrence”, not “convergence”. A6 removed all Tier B observations: exactly one gene still recurs across studies, *sbmA*, with three direct events (Spohn PR39_5 short deletion of 1 bp at 392,493; Bac7 strain B1 IC3-like IS insertion of 1.4 kb; strain B2 SNP). However, the eight Spohn lines carrying *sbmA* through large deletions coalesce to three structural events that share the start breakpoint 388,423: $\Delta 5,466$ (PR39_2), $\Delta 7,615$ (BAC5_7) and $\Delta 8,821$ bp (six lines). A further $\Delta 11,471$ bp deletion at 275,642 appears in 11 lines across five AMPs. Sixty-five of 93 within-study recurrences are driven solely by shared large deletions and count once biologically.

3.4 Orthology and annotation (A7)

WaaY (P27240, heptose II kinase) is not functionally equivalent to WaaP (P25741, heptose I kinase, EC 2.7.1.235); counts are not pooled. The orthology of *S. maltophilia* *SbmA* could not be resolved: the per-organism KEGG control failed to discriminate, so empty results are inconclusive. The *sbmA* claim is therefore *E. coli* only.

3.5 Mutation-opportunity null (A7b)

Against a length-proportional null over a pinned 4,318-CDS MG1655 universe, the observed number of genes hit in at least two studies was 1 (sbmA), with null mean 0.405 and $p = 0.339$ (100,000 draws, seed 20261007). The locked analysis wrongly placed Blanco (*S. maltophilia*) in the *E. coli* universe; the *E. coli*-only sensitivity run gave $p = 0.116$. The sbmA-specific tail probability (0.00011) is post hoc and not a test. Result: sbmA recurrence is not shown to exceed mutation opportunity.

3.6 IS-element association of deletion breakpoints (A8)

Four of nine distinct Spohn large-deletion breakpoints lie within 200 bp of an IS element in the BW25113 reference (null mean 0.131; 0 of 100,000 draws ≥ 4 ; $p < 10^{-5}$). They come from only three independent loci: the $\Delta 11,471$ bp deletion is flanked by IS1B and IS1C, the 388,423 start sits at IS3B, and the 3,645,395 end lies inside IS5T. A locus-level analytic check gave about 8×10^{-5} . This supports reading the sbmA block lines as breakpoint recurrence rather than eight independent selections.

3.7 Salmonella candidate concordance (A9)

Locked candidates waaY and phoP were drawn from the *Salmonella* LL-37 passaging study. Identities to the *E. coli* orthologs were 69.8% (WaaY, 162 of 232) and 92.9% (PhoP, 208 of 224), using sequences from strains F98 and ATCC 14028. In the 121 mapped Spohn Tier A events, T (candidates hit) was 1: waaY hit, phoP not. The locked rule required $T=2$ with $p < 0.05$; null $P(T \geq 1) = 0.0406$. **The rule was not met** and no concordance claim is made. Descriptively, waaY has 8 Tier A events in 8 lines and basS has 9 events in 9 lines. A9’s own text refers to “8 of 120 lines”; the 120 figure is **superseded** by Section 3.8 and should not be quoted.

3.8 Line dependence of waaY, basS and sbmA-deletion lines (A10, corrected)

A10 tested whether lines carrying waaY hits (W, 8 lines), basS hits (B, 9) and sbmA deletions (S, 8) overlap more than expected. The lock assumed 120 lines. A first run used 60 lines and reported $W \cap S = 5$ with stratified $p = 0.0046$ (**withdrawn**: wrong denominator, not an evidential result). The pinned Supplementary Data 5 states 38 sequenced lines, and the six adopted AMPs have 28; the correct universe is 28 lines with per-AMP sizes 4/5/4/5/5/5. The first-run result is **withdrawn**. With 28 lines, $W \cap S = 5$ has an AMP-stratified permutation $p = 0.601$ (null mean 4.60); $B \cap S = 2$, $p = 0.601$; $W \cap B = 3$, $p = 0.331$ (Table 1, Figure 1). Under the locked rule no dependence beyond AMP exposure is shown, and the data cannot separate shared-line dependence from AMP exposure because BAC5 and PR39 strata hold 4 and 5 lines. The 5-of-8 overlap of waaY and sbmA-deletion lines remains a descriptive caveat.

Table 1: A10 line overlaps under the corrected 28-line universe (seed 20261010, 100,000 permutations). The first-run values (60 lines) are withdrawn.

Overlap	Observed	Hypergeometric p	Stratified null mean	Stratified p
$W \cap S$	5	0.022	4.60	0.601
$B \cap S$	2	0.830	1.60	0.601
$W \cap B$	3	0.516	2.10	0.331

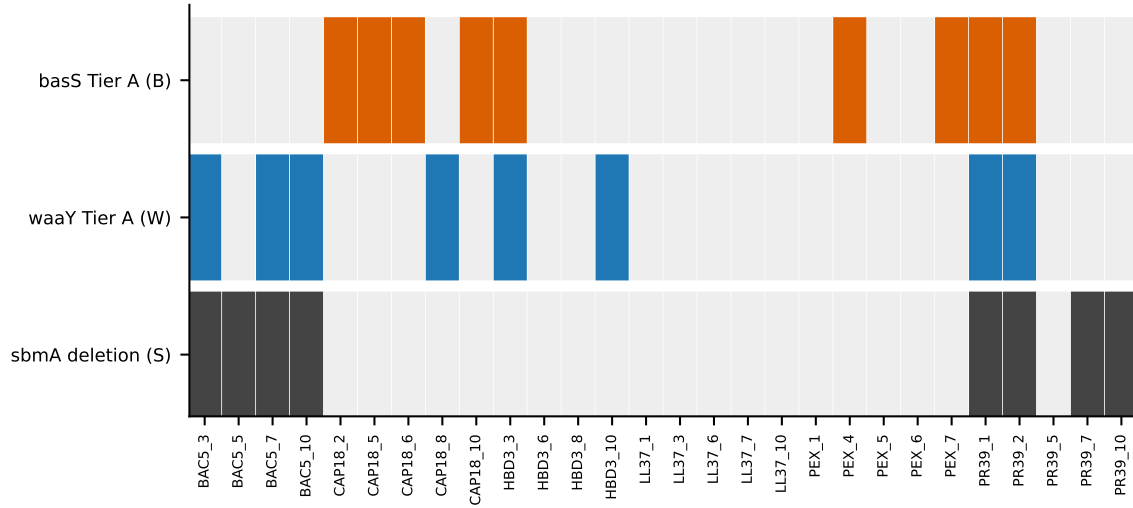


Figure 1: Membership of the 28 sequenced adopted-AMP Spohn lines in the sbmA-deletion set (S), the waaY Tier A set (W) and the basS Tier A set (B). Data: results/line_dependence_a10_corrected_universe28.json.

3.9 The basS and pmrB mapping (A11)

E. coli BasS (P30844, 363 aa) and *S. typhimurium* LT2 PmrB (P36557, 356 aa; UniProt names basS, parB, pmrB) share 306 identities (84.3% of the longer sequence; length ratio 0.981), meeting the locked name-level rule (at least 70% and ratio at least 0.9). A simple alignment (+1/−1/−2) was used because BLOSUM62 was unavailable; 84.3% is far from the boundary. Sequence similarity supports orthology, not functional equivalence.

3.10 Recurrence outside sbmA-deletion lines (A12, A14)

A12 excluded the eight sbmA-deletion lines and intergenic and Tier B events, leaving 20 lines (CAP18 5, PEX 5, LL-37 5, HBD3 4, PR39 1) and 64 mapped Tier A events. The statistic T counted (AMP, gene) pairs with Tier A hits in at least three same-AMP lines. Observed T was 4 (Table 2). The null redraw each line’s number of distinct genes from the 4,318 CDS in proportion to length (null mean 1.2×10^{-4} ; 0 of 100,000 reps ≥ 4), so the Monte Carlo tail estimate is $< 10^{-5}$ under that specified null (the 100,000-draw resolution cannot distinguish smaller values). The first run counted yejK and yejL (one intergenic event pair) in violation of the lock (T=6, **withdrawn**: it violated the locked intergenic exclusion); it was corrected and both files are kept. A14 resolved the seven excluded events whose gene names are old synonyms (yejM→lapC, yciM→lapB, yebS→letA, yebT→letB, ybiH→cecR; each a unique reviewed UniProt K-12 entry) and the statistic was unchanged (71 mapped events, T=4, 0 of 100,000 draws reaching 4). A14 does not independently validate A12.

This is a statement about the specified simulation only; it is not evidence that the recurrence has probability $< 10^{-5}$ under a biologically realistic null. The null treats genes as independent draws weighted by length. It ignores insertion-sequence hotspots, linked or co-transcribed genes and mutation-rate heterogeneity, so the extreme tail value should not be read as a precise probability. basS and basR are one two-component system, and lptC and wzzE share the line LL37_3. Lines within an AMP are not independent, and strata

Table 2: Within-AMP Tier A recurrence outside sbmA-deletion lines (A12; A14 unchanged).

AMP	Gene	Lines hit
CAP18	basS	4
LL-37	lptC	3
LL-37	wzzE	3
HBD3	basR	3

hold only four or five lines.

3.11 A second organism, exploratory (A13, A13b)

A13 applied the same statistic to the 16 Blanco evolved lines against 4,324 *S. maltophilia* D457 CDS (NC_017671.1). As locked, name-only mapping resolved only 5 of 16 events (RefSeq carries no sspB or mraW symbols), giving $T=0$ and $p = 1.0$: **not testable as locked**, an annotation gap and not a finding. A13b added two mappings chosen after seeing which genes recurred (mraW→rsmH; sspB→the single CDS annotated “ClpXP protease specificity-enhancing factor”). Then 14 of 16 events mapped, $T=2$ (PR-39: 6 of 8 lines; LL-37 rsmH: 3 of 8), and 0 of 100,000 reps reached 2 ($p < 10^{-5}$). Because the mapping was chosen post hoc, this is **exploratory** and is not counted as a clean replication. It gives an exploratory indication that within-AMP recurrence can also occur in this dataset; it cannot establish replication of the *E. coli* observation, and neither T nor its tail estimate has confirmatory status. The recurring genes differ from those in *E. coli*. Blanco lines come from a shared parent, which can inflate recurrence. Bac7 2023 (3 strains) was not testable.

3.12 The 09d gate search (A15)

Before searching, criteria were locked: a PubMed-indexed paper independent of Spohn 2019 with a defined perturbation of basS, basR, lptC or wzzE (or the *Salmonella* ortholog for basS) and a measured change in MIC or killing for the same AMP or family, with the direction stated. Five PubMed queries returned 20 distinct PMIDs, screened by abstract, with one full text read. The results:

- **basS/pmrB: partial.** Lofton et al. 2013 (PMID 23894360) found pmrB mutations (H164R in an LL-37 isolate; R13H in a wheat-germ-histone isolate) in passaged *Salmonella* lineages. The reconstituted R13H gave only a small resistance increase, to wheat germ histones; the authors report the assay was not sensitive enough to show an effect and that pmrB had negligible effect in combinations. An LL-37 effect of pmrB is therefore not demonstrated.
- **waaY:** the same paper shows a reconstituted waaY deletion gives about 2 to 4 logs higher survival against all three peptides including LL-37. This is external functional evidence from *Salmonella*, but it does not validate the *E. coli* recurrence result, waaY is not among the surviving genes, and it does not rescue A9.
- **basR/pmrA: partial.** PmrA-PmrB governs resistance to polymyxin and other cationic peptides in *Salmonella* (PMIDs 8955307, 11035717), but one report states PmrA-PmrB was not responsible for NP-1 defensin sensitivity, and another finds peptide-specific pmrA effects (PMID 29267354). No hBD3 measurement was found.
- **lptC:** no qualifying record. **wzzE:** no hit.

Mass spectrometry was not applicable because no peptide is named. The gate is **unmet**.

4 Discussion

The lane’s original objective, predicting resistance emergence from sequence features, failed its locked tests on the available data, and the data structure explains why: the labels are few, nested and partly censored. The pivot to mutation recurrence produced one reproducible descriptive pattern within the current *E. coli* ledger: four AMP–gene pairs recur in at least three same-AMP lines after the specified exclusions and corrections, more often than the specified length-weighted redraw procedure generates. That null does not capture mutation hotspots, linkage, co-transcription or dependence among lines (lines within an AMP are not statistically independent), so the pattern is not shown to exceed a biologically realistic mutation-opportunity process. Four pairs in 20 lines, drawn from four AMP strata of 4 to 5 lines each, is a small base, and the finding is hypothesis-generating. The affected genes have functions related to LPS, outer-membrane biology or envelope-associated regulation (a two-component system, an LPS transport protein and an O-antigen chain length regulator); whether those functions explain the recurrence is not tested here.

Three earlier readings should not be repeated. First, *sbmA* recurrence cannot be called convergent selection: it does not beat a mutation-opportunity null, and most of its block evidence is breakpoint driven by IS elements. Second, *waaY* is not shown to depend on the *sbmA*-deletion lines; the briefly reported $p = 0.0046$ (withdrawn) came from a wrong denominator. Third, *waaY/phoP* concordance with *Salmonella* is not established. Lofton et al. 2013 provides external evidence that some LPS-associated and two-component-system loci recur or affect peptide resistance in *Salmonella*. This is biologically consistent with, but does not validate, the broader reading of the *E. coli* pattern: it concerns different genes in another organism, and the study had already been mined by this lane (rated partial in the A2 mining log), so it is not an independent replication.

5 Limitations

(i) The null model is length-weighted and ignores mutation hotspots and linkage. (ii) Spohn strata hold 4 to 5 lines per AMP, and lines within an AMP are not independent. (iii) A13b rests on a post hoc mapping, and Blanco lines share a parent. (iv) Several analyses were run after the analyst had seen related counts; each lock states that exposure. (v) Reference-genome features were used, not each line’s own assembly. (vi) The PubMed search was bounded: abstracts of 20 records and one full text. (vii) Several first-run results were wrong and were corrected; the originals are kept in the repository.

6 Claim state and closing condition

Survives as a descriptive pattern: four AMP–gene pairs recur in at least three same-AMP lines outside the *sbmA*-deletion set, more extreme than the specified length-weighted redraw null (Section 3.10). This does not establish recurrence beyond a biologically realistic mutation-opportunity null. The second-organism result (A13b) is exploratory, and Lofton 2013 is external, partial evidence and not an independent replication.

Does not survive: the 09d gate, *sbmA* recurrence exceeding mutation opportunity, *waaY* concordance, and *waaY* dependence on *sbmA*-deletion lines. No candidate peptide is nominated and no discovery is claimed. One route to closing the gate is a targeted functional test, for example a *basS*, *lptC* or *wzzE* allele measured

against LL-37 or CAP18 MIC or killing in *E. coli* BW25113, with a prespecified direction; no such test has been run.

7 A16: conditional permutation of the A12 pattern

A16 (lock 16f8156, run as locked, not blind to the data) re-tested $T=4$ against a null that keeps each line's number of genes and each gene's total number of line hits fixed (curveball, 100,000 samples). 240 of 100,000 samples reached $T=4$ (null mean 0.96, 99th percentile 3), so $p = 0.0025$; this is the better figure than the A12 length-weighted tail estimate, which overstated strength. A locked sensitivity that merges same-strand genes within 1,000 bp into neighborhoods gave $T=5$ against a null 95th percentile of 5 ($p = 0.078$): the pattern does not stand out once linked genes are merged. The honest reading is a modest margin-conditional excess that is not robust to linkage, still hypothesis-generating, with the 09d gate unmet. Details: docs/A16_CONDITIONAL_PERMUTATION_RESULT_2026-10-08.md.

8 Length of this document

The body of this document (Sections 1 to 8, excluding the appendix, references and front matter) is about **7 pages** (it ends on page 8 of a 9-page PDF; page 9 holds the appendices). The repository requires a 50+ page paper; that requirement is **unmet**. The evidence committed so far supports about this much honest text, and the document has not been padded.

A Disclosed deviations

Table 3: Deviations disclosed in the repository, with their handling.

Analysis	Deviation	Handling
A1	Assumed sequence-level benchmark not supported; AUROC dropped	Disclosed in A1.2
A7b	Blanco (<i>S. maltophilia</i>) placed in the <i>E. coli</i> universe	<i>E. coli</i> -only sensitivity $p = 0.116$ reported
A10	Lock said 120 lines; first run used 60; correct universe is 28	First run withdrawn and kept; corrected $p = 0.601$
A11	BLOSUM62 not available; empty first pin replaced	Simple alignment; margin large
A12	First run omitted the intergenic exclusion (T=6)	Corrected (T=4); both files kept
A13	Name-only mapping resolved 5 of 16 events	Reported not testable
A13b	Mapping chosen after seeing recurring genes	Labelled exploratory
A15	Search limited to abstracts plus one full text	Stated in Section 3.12

B Reproducibility

Scripts are in `scripts/`; results in `results/`; locks and result documents in `docs/`. Lock commits: A7 036e075; A7b 9e50efd; A8 d60c916; A9 35256e1; A10 7d0699e; A11 458119f; A12 57e242c; A13 36fe1c2; A13b f27e8cf; A14 81ffe43; A15 9504af1. Unit tests live in `tests/`. Primary sources: Spohn et al. 2019 (Supplementary Data 5); Blanco et al. 2020; Bac7 2023 (supplement); Lofton et al. 2013, PLoS One 8(7):e68875 (PMID 23894360); PMIDs cited in Section 3.12.